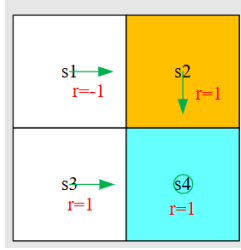


§ Lec III

Bellman Optimality Equation (BOE)

1. Motivating Example



For the given policy π , we can write out the Bellman equations:

$$v_{\pi}(s_1) = -1 + \gamma v_{\pi}(s_2),$$

$$v_{\pi}(s_2) = +1 + \gamma v_{\pi}(s_4),$$

$$v_{\pi}(s_3) = +1 + \gamma v_{\pi}(s_4),$$

$$v_{\pi}(s_4) = +1 + \gamma v_{\pi}(s_4).$$

Let $\gamma = 0.9$, the *state values* are :

$$v_{\pi}(s_4) = v_{\pi}(s_3) = v_{\pi}(s_2) = 10, v_{\pi}(s_1) = 8.$$

For *action values*: consider s_1 :

$$q_{\pi}(s_1, a_1) = -1 + \gamma v_{\pi}(s_1) = 6.2,$$

$$q_{\pi}(s_1, a_2) = -1 + \gamma v_{\pi}(s_2) = 8,$$

$$q_{\pi}(s_1, a_3) = 0 + \gamma v_{\pi}(s_3) = 9,$$

$$q_{\pi}(s_1, a_4) = -1 + \gamma v_{\pi}(s_1) = 6.2,$$

$$q_{\pi}(s_1, a_5) = 0 + \gamma v_{\pi}(s_1) = 7.2.$$

Q: However, this current policy is not good, how can we improve it ?

A: By using action values !

The current policy $\pi(a|s_1)$ is:

$$\pi(a|s_1) = \begin{cases} 1 & a = a_2 \\ 0 & a \neq a_2 \end{cases}$$

Observe the action values that we obtained just now:

$$q_{\pi}(s_1, a_1) = 6.2, \quad q_{\pi}(s_1, a_2) = 8, \quad q_{\pi}(s_1, a_3) = 9,$$

$$q_{\pi}(s_1, a_4) = 6.2, \quad q_{\pi}(s_1, a_5) = 7.2.$$

What if we select the greatest action value? (this time a_3) Then, a new policy is obtained:

$$\pi_{\text{new}}(a|s_1) = \begin{cases} 1 & a = a^* \\ 0 & a \neq a^* \end{cases}$$

where $a^* = \operatorname{argmax}_a q_{\pi}(s_1, a) = a_3$.

(the new policy will always choose the greatest action)

Why choosing the greatest action value could improve the policy?

Intuition: actions with greater value is better because we will accumulate more returns.

2. Optimal Policy

The state value could be used to evaluate if a policy is good or not: if

$$v_{\pi_1}(s) \geq v_{\pi_2}(s) \quad \text{for all } s \in \mathcal{S}$$

then π_1 is “better” than π_2 .

• Definition

A policy π^* is optimal if $v_{\pi^*}(s) \geq v_{\pi(s)}$ for all s and for any other policy π .

The definition leads to many questions:

- ① Does the optimal policy exist?
- ② Is the optimal policy unique?
- ③ Is the optimal policy stochastic or deterministic?
- ④ How to obtain the optimal policy?

To answer these questions, we study the Bellman optimality equation.

• Bellman optimality equation (BOE)

BOE (elementwise form):

$$v(s) = \max_{\pi} \sum_a \pi(a|s) \left(\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v(s') \right), \forall s \in \mathcal{S}$$

$$= \max_{\pi} \sum_a \pi(a|s) q(s, a), s \in \mathcal{S}$$

It's just Bellman equation with an optimal π !

- ① $p(r|s, a), p(s'|s, a)$ are known.
- ② $v(s), v(s')$ are unknown and to be calculated.
- ③ Is $\pi(s)$ known or unknown?

BOE (matrix-vector form):

$$v = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$$

$$[r_{\pi}]_s \triangleq \sum_a \pi(a|s) \sum_r p(r|s, a)r,$$

$$[P_{\pi}]_{s,s'} = p(s'|s) \triangleq \sum_a \pi(a|s) \sum_{s'} p(s'|s, a)$$

Here $\max_{\pi}$ is performed elementwise:

$$\max_{\pi} v = \begin{pmatrix} \max_{\pi} v_1 \\ \max_{\pi} v_2 \\ \max_{\pi} v_3 \end{pmatrix}$$

Consider the matrix-vector form, we have to solve *two* unknowns(v, π) from just *one* equation.

• ▶ Fix $v'(s)$ first and solve π :

$$v(s) = \max_{\pi} \sum_a \pi(a|s) \left(\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v(s') \right), \forall s \in \mathcal{S}$$

$$= \max_{\pi} \sum_a \pi(a|s) q(s, a)$$

Example (How to solve $\max_{\pi} \sum_a \pi(a|s) q(s, a)$)

Suppose $q_1, q_2, q_3 \in \mathbb{R}$ are given. Find c_1^*, c_2^*, c_3^* solving

$$\max_{c_1, c_2, c_3} c_1 q_1 + c_2 q_2 + c_3 q_3.$$

where $c_1 + c_2 + c_3 = 1$ and $c_1, c_2, c_3 \geq 0$.

Without loss of generality, suppose $q_3 \geq q_1, q_2$. Then, the optimal solution is $c_3^* = 1$ and $c_1^* = c_2^* = 0$. That is because for any c_1, c_2, c_3

$$q_3 = (c_1 + c_2 + c_3)q_3 = c_1 q_3 + c_2 q_3 + c_3 q_3 \geq c_1 q_1 + c_2 q_2 + c_3 q_3.$$

Inspired by the above example, considering that $\sum_a \pi(a|s) = 1$, we have

$$\max_{\pi} \sum_a \pi(a|s) q(s, a) = \max_{a \in \mathcal{A}(s)} q(s, a),$$

$$\pi(a|s) = \begin{cases} 1 & \text{if } a = a^* \\ 0 & \text{if } a \neq a^* \end{cases}$$

$$a^* = \arg \max_a q(s, a)$$

where the optimality is achieved as shown above.(note: the π we get is a function of v)

• ▶ Solve the BOE

The BOE is $v = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$. Let

$$f(v) := \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$$

Then, the Bellman optimality equation becomes(π is also a function of v)

$$v = f(v)$$

where

$$[f(v)]_s = \max_{\pi} \sum_a \pi(a|s) q(s, a), s \in \mathcal{S}$$

Next, how to solve the equation?

Preliminaries:

• **Some concepts:**

① **Fixed point:** $x \in X$ is a fixed point of $f : X \rightarrow X$ if

$$f(x) = x$$

② **Contraction mapping** (or contractive function): f is a contraction mapping if

$$\|f(x_1) - f(x_2)\| \leq \gamma \|x_1 - x_2\|$$

where $\gamma \in (0, 1)$.

γ must be strictly less than 1 so that many limits such as $\gamma^k \rightarrow 0$ as $k \rightarrow \infty$ hold.

Here $\|\cdot\|$ can be any vector norm.

• **Contraction mapping theorem:**

For any equation that has the form of $x = f(x)$, if f is a contraction mapping, then

① **Existence:** there exists a fixed point x^* satisfying $f(x^*) = x^*$.

② **Uniqueness:** The fixed point x^* is unique.

③ **Algorithm**(iterative): Consider a sequence $\{x_k\}$ where $x_{k+1} = f(x_k)$, then $x_k \rightarrow x^*$ as $k \rightarrow \infty$. Moreover, the convergence rate is exponentially fast.

there always exists a solution v^* and the solution is unique. The solution could be solved iteratively by

$$v_{k+1} = f(v_k) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v_k)$$

This sequence $\{v_k\}$ converges to v^* exponentially fast given any *initial guess* v_0 . The convergence rate is determined by γ .

• **Policy optimality**

Suppose v^* is the solution to the BOE. It satisfies

$$v^* = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v^*)$$

Suppose

$$\pi^* = \arg \max_{\pi} (r_{\pi} + \gamma P_{\pi} v^*)$$

Then

$$v^* = r_{\pi^*} + \gamma P_{\pi^*} v^*$$

Therefore, π^* is a policy and $v^* = v_{\pi^*}$ is the corresponding state value.

(Note: we use the first guessed v_0 to iteratively converge to the final v^* and extract π^* greedily from v^*)

Let's go back to the former BOE:

$$v = f(v) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$$

$f(v)$ is a contraction mapping satisfying

$$\|f(v_1) - f(v_2)\| \leq \gamma \|v_1 - v_2\|$$

where γ is the discount rate!

Applying the contraction mapping theorem gives the following results.

For the BOE

$$v = f(v) = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$$

Theorem (Policy Optimality)

Suppose that v^* is the unique solution to $v = \max_{\pi} (r_{\pi} + \gamma P_{\pi} v)$, and v_{π} is the state value function satisfying $v_{\pi} = r_{\pi} + \gamma P_{\pi} v_{\pi}$ for any given policy π (that's a Bellman equation), then it can be proved that

$$v^* \geq v_{\pi}, \forall \pi$$

$$(v^* = v_{\pi^*})$$

Note that π is not necessarily unique ! (but v^* is unique !)

What does π^* look like?

Theorem (Greedy Optimal Policy)

For any $s \in \mathcal{S}$, the **deterministic greedy policy**

$$\pi^*(a|s) = \begin{cases} 1 & a = a^*(s) \\ 0 & a \neq a^*(s) \end{cases}$$

is an optimal policy solving the BOE. Here,

$$a^*(s) = \arg \max_a q^*(a, s),$$

(that is, π^* always takes the action $a^*(s)$ that has the greatest action value $q^*(a, s)$)

where

$$q^*(s, a) := \sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v^*(s').$$

- How changing r and γ can change the optimal policy ?

- ▶ If we lower γ , the optimal policy will become more short-sighted!

When γ is set to 0, the optimal policy will always choose the action that has the greatest immediate reward!

- ▶ If we lower $r_{\text{forbidden}}$, the optimal policy will tend to avoid the forbidden area.

Also, if we change $r \rightarrow ar + b$, the optimal policy will remain the same. (what matters is the relative values)

Proof:

$$\pi^*(s) = \arg \max_{\pi} \sum_a \pi(a|s) \underbrace{\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v^*(s')}_{q^*(s, a)}$$

3. Analyzing optimal policies

- What factors determine the optimal policy ?

It can be clearly seen from the BOE

$$v(s) = \max_{\pi} \sum_a \pi(a|s) \left(\sum_r p(r|s, a)r + \gamma \sum_{s'} p(s'|s, a)v(s') \right)$$

that there are three factors:

- ① Reward design: r
- ② System model: $p(s'|s, a), p(r|s, a)$
- ③ Discount rate: γ

$v(s), v(s'), \pi(a|s)$ are unknowns to be calculated

Theorem (Optimal Policy Invariance)

Consider a Markov decision process with $v^* \in \mathbb{R}^{|\mathcal{S}|}$ as the optimal state value. If every reward $r \rightarrow ar + b$, ($a, b \in \mathbb{R}$ and $a \neq 0$), then the corresponding optimal state value v' is

$$v' = av^* + \frac{b}{1 - \gamma} \mathbf{1}$$

where $\gamma \in (0, 1)$ is the discount rate and $\mathbf{1} = [1, \dots, 1]^T$. Consequently, the optimal policies are invariant to this transformation.

Meaningless detour: the discount rate gives penalty for taking more steps.